

```
data["NumProjects"] = 378

// JSON must have a BOM, so turn it off with empty last argument
writeStringToFile(data.asString, "D:\tnt.json", false, "")
```

See also

[readStringFromFile \[SimTalk\]](#)

[Program \[SimTalk\]](#)

Query the Data Type of a Variable of Data Type any

Query the Data Type of a Variable of Data Type Any

SimTalk provides the functions listed in the table of contents to the left for querying the data type of a variable of data type *Any*.

The function expects a parameter whose data type *Plant Simulation* compares.

See also

[Any \[SimTalk\] - data type](#)

Return Value

The return value has the data type *boolean*.

getSimTalkTypename [SimTalk]

Returns the name of the data type of an arbitrary value.

Type

Function

Syntax

```
getSimTalkTypename(Value:any) → string
```

Return Value

The return value for the data types *table/stack/queue* returns these strings:

Data Type [SimTalk]	Return Value [SimTalk]
<i>table</i> [SimTalk]	<i>table</i> [SimTalk]
<i>stack</i> [SimTalk]	<i>stack</i> [SimTalk]
<i>queue</i> [SimTalk]	<i>queue</i> [SimTalk]

For local variables of data type *any*, which were assigned a special object, for example a sensor, *getSimTalkTypename* returns **sensor**, **lane**, **storage place**, **3D object**, or **3D animation** depending on the type of the special object.

If the passed value has an *Arraydata* type, *getSimTalkTypename* returns an *array*.

Examples

```
print getSimTalkTypename(Conveyor.sensorID(1)) // sensor
print getSimTalkTypename(Conveyor._3D)        // 3D object
print getSimTalkTypename(TwoLaneTrack.A)        // lane
print getSimTalkTypename(Store[1,1])            // storage place
```

```
// compute the character count
param value: any -> integer

switch getSimTalkTypename(value)
case "integer"
  if value > 0
    return log10(value) + 1
  elseif value < 0
    return log10(-value) + 2
  else
```

```
        return 1;
    end
    case "string"
        return strlen(value)
    case "object"
        return strlen(obj_to_str(value))
    case "boolean"
        return when value then /*true*/4 else /*false*/5
    else
        return -1 // error
    end
```

See also

[Any \[SimTalk\] - data type](#)

[Array \[SimTalk\] - data type](#)

isAcceleration [SimTalk]

Returns if the designated argument is of data type *Acceleration* (true) or not (false).

Type

Function

Syntax

```
isAcceleration(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isAcceleration(arg)
  print "acceleration = , arg"
end
```

See also

[Any \[SimTalk\] - data type](#)

[Acceleration \[SimTalk\] - data type](#)

isArray [SimTalk]

Returns if the designated argument has an *Array* data type (true) or not (false).

Type

Function

Syntax

```
isArray(Argument:Value) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
var a:any[] := [[1,2,3], 42]
print "a = ", a
print "a[1] = ", a[1]
print "a[2] = ", a[2]

print isArray(a)      // true
print isArray(a[1])   // true
print isArray(a[2])   // false
```

```
var j:json
j["a"] := [1,2,3]
print isArray(j["a"]) -- returns true
```

See also

[Array \[SimTalk\] - data type](#)

isBoolean [SimTalk]

Returns if the designated argument is of data type *Boolean* (true) or not (false).

Type

Function

Syntax

```
isBoolean(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isBoolean(arg)
    print "Boolean = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Boolean \[SimTalk\] - data type](#)

isDate [SimTalk]

Returns if the designated argument is of data type *Date* (true) or not (false).

Type

Function

Syntax

```
isDate(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isDate(arg)
    print "Date = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Date \[SimTalk\] - data type](#)

isDatetime [SimTalk]

Returns if the argument designated is of data type *DateTime* (true) or not (false).

Type

Function

Syntax

```
isDatetime(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isDatetime(arg)
  print "Datetime = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[DateTime \[SimTalk\] - data type](#)

isInteger [SimTalk]

Returns if the designated argument is of data type *Integer* (true) or not (false).

Type

Function

Syntax

```
isInteger(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isInteger(arg)
    print "Integer = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Integer \[SimTalk\] - data type](#)

isJson [SimTalk]

Returns if the designated argument is of data type *JSON* (true) or not (false).

Type

Function

Syntax

```
isJson(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isJson(arg)
    print "json = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[JSON \[SimTalk\] - data type](#)

isLength [SimTalk]

Returns if the designated argument is of data type *Length* (true) or not (false).

Type

Function

Syntax

```
isLength(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isLength(arg)
    print "Length = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Length \[SimTalk\] - data type](#)

isList [SimTalk]

Returns if the designated argument is of data type *List* (true) or not (false).

Syntax

```
isList(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Note

The value VOID does not have a data type as local variables of data types *object*, *table*, and *list* can all take the value void. For this reason a local variable of data type *any* does not take a data type when you assign void to it.

Example

```
var a: any := void
print isObject(a) // prints false
print a = void    // prints true

var b : any := 123
print b = void    // prints false
```

See also

[Any \[SimTalk\] - data type](#)

[List \[SimTalk\] - data type](#)

isListRange [SimTalk]

Returns if the designated argument is a list range (true) or not (false).

Type

Function

Syntax

```
isListRange(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isListRange(arg)           // something like {1,1}..{2,*}
    print "listRange = ", arg // prints listRange = {1,1}..{2,*}
end
```

See also

[Any \[SimTalk\] - data type](#)

isObject [SimTalk]

Returns if the designated argument is of data type *Object* (true) or not (false).

Type

Function

Syntax

```
isObject(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Note

For *isObject*, *isTable*, and *isList* this applies: The value VOID does not have a data type as local variables of data types *object*, *table*, and *list* can all take the value void. For this reason a local variable of data type *any* does not take a data type when you assign void to it. If the function *isObject* is called with such a variable, the function returns the result false.

Example

```
var a: any := void
print isObject(a) // prints false
print a = void    // prints true

var b : any := 123
print b = void    // prints false
```

See also

[Any \[SimTalk\] - data type](#)

[Object \[SimTalk\] - data type](#)

isQueue [SimTalk]

Returns if the designated argument is of data type *Queue* (true) or not (false).

Type

Function

Syntax

```
isQueue(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isQueue(arg)
  print "Queue = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Queue \[SimTalk\] - data type](#)

isReal [SimTalk]

Returns if the designated argument is of data type *Real* (true) or not (false).

Type

Function

Syntax

```
isReal(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isReal(arg)
  print "Real = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Real \[SimTalk\] - data type](#)

isSpeed [SimTalk]

Returns if the designated argument is of data type *Speed* (true) or not (false).

Type

Function

Syntax

```
isSpeed(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isSpeed(arg)
  print "Speed = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Speed \[SimTalk\] - data type](#)

isStack [SimTalk]

Returns if the designated argument is of data type *Stack* (true) or not (false).

Type

Function

Syntax

```
isStack(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isStack(arg)
  print "Stack = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Stack \[SimTalk\] - data type](#)

isString [SimTalk]

Returns if the designated argument is of data type *String* (true) or not (false).

Type

Function

Syntax

```
isString(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isString(arg)
    print "String = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[String \[SimTalk\] - data type](#)

isTable [SimTalk]

Returns if the designated argument is of data type *Table* (true) or not (false).

Type

Function

Syntax

```
isTable(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Note

The value VOID does not have a data type as local variables of data types *object*, *table*, and *list* can all take the value void. For this reason a local variable of data type *any* does not take a data type when you assign void to it.

Example

```
param arg: any
if isTable(arg)
    print "Table = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Table \[SimTalk\] - data type](#)

isTime [SimTalk]

Returns if the designated argument is of data type *Time* (true) or not (false).

Type

Function

Syntax

```
isTime(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any  
if isTime(arg)  
    print "Time = ", arg  
end
```

See also

[Any \[SimTalk\] - data type](#)

[Time \[SimTalk\] - data type](#)

isWeight [SimTalk]

Returns if the designated argument is of data type *Weight* (true) or not (false).

Type

Function

Syntax

```
isWeight(Argument:any) → boolean
```

Parameter

The parameter *Argument* of data type *any* designates the argument to be checked.

Return Value

The return value has the data type *boolean*.

Example

```
param arg: any
if isweight(arg)
  print "weight = ", arg
end
```

See also

[Any \[SimTalk\] - data type](#)

[Weight \[SimTalk\] - data type](#)

Input Functions

Input Functions

SimTalk provides the functions listed in the table of contents to the left for requesting input from the user.

When you create the simulation model certain data will not be known. You might, for example, ask the user to enter the **capacity** of a *Store*, which you inserted into a *Frame*.